QUBO Formulation: Robot Arm Path Planning

Robot Arm Path Planning

Problem Statement. Given a discrete time horizon T and a sequence of state sets $\mathcal{S}_t$ for each time step $t \in \{0, \ldots, T-1\}$, the Robot Arm Path Planning problem seeks a sequence of states $(s_0, s_1, \ldots, s_{T-1})$ with $s_t \in \mathcal{S}_t$ that minimizes the sum of state-dependent obstacle costs and transition-dependent motion costs, subject to validity constraints on consecutive state pairs and fixed initial and final states.

Decision Variables. For each time step $t \in \{0, \ldots, T-1\}$ and state $s \in \mathcal{S}_t$, introduce a binary variable $x_{t,s} \in \{0, 1\}$. The variable $x_{t,s}$ equals 1 if and only if state s is selected at time step t, and 0 otherwise.

QUBO Derivation. Step 1 — Original Objective. The problem is to minimize the total obstacle cost plus the total motion cost:

$$\min \quad \sum_{t=0}^{T-1} \sum_{s \in \mathcal{S}_t} C_{\text{obs}}(s) x_{t,s} + \sum_{t=0}^{T-2} \sum_{s \in \mathcal{S}_t} \sum_{s' \in \mathcal{S}_{t+1}} C_{\text{mot}}(s, s') x_{t,s} x_{t+1,s'}. \tag{1}$$

Step 2 — Constraints. The feasible region is defined by:

$$\text{(C1)} \quad \sum_{s \in \mathcal{S}_t} x_{t,s} = 1, \quad \forall t \in \{0, \ldots, T-1\}, \tag{2}$$

$$\text{(C2)} \quad x_{t,s} + x_{t+1,s'} \leq 1, \quad \forall \text{ invalid transitions } (s, s') \text{ between } t \text{ and } t+1, \tag{3}$$

$$\text{(C3)} \quad x_{0,s_{\text{start}}} = 1, \tag{4}$$

$$\text{(C4)} \quad x_{T-1,s_{\text{goal}}} = 1. \tag{5}$$

Step 3 — Penalty Conversion. We convert each constraint into a quadratic penalty term added to the objective. For the equality constraint (C1), we add $\lambda_{\text{eq}} \left(\sum_{s \in \mathcal{S}_t} x_{t,s} - 1 \right)^2$.

Using the idempotent property $x_{t,s}^2 = x_{t,s}$ for binary variables, we expand:

$$\begin{aligned}\left(\sum_{s\in\mathcal{S}_t} x_{t,s} - 1\right)^2 &= \sum_{s\in\mathcal{S}_t} x_{t,s}^2 + 2\sum_{\substack{s,s'\in\mathcal{S}_t\\ s<s'}} x_{t,s}x_{t,s'} - 2\sum_{s\in\mathcal{S}_t} x_{t,s} + 1 \\ &= -\sum_{s\in\mathcal{S}_t} x_{t,s} + 2\sum_{\substack{s,s'\in\mathcal{S}_t\\ s<s'}} x_{t,s}x_{t,s'} + 1. \end{aligned} \tag{6}$$

For the inequality constraint (C2), since $x_{t,s}, x_{t+1,s'} \in \{0,1\}$, the condition $x_{t,s} + x_{t+1,s'} \leq 1$ is violated if and only if both variables equal 1. Thus, the penalty is simply $\lambda_{\text{ineq}} x_{t,s}x_{t+1,s'}$. For the fixed start and goal constraints (C3) and (C4), we use $\lambda_{\text{start}}(x_{0,s_{\text{start}}}-1)^2 = \lambda_{\text{start}}(-x_{0,s_{\text{start}}}+1)$ and $\lambda_{\text{goal}}(x_{T-1,s_{\text{goal}}} - 1)^2 = \lambda_{\text{goal}}(-x_{T-1,s_{\text{goal}}} + 1)$.

Step 4 — Combined Hamiltonian. Summing the objective and all penalty contributions yields the QUBO Hamiltonian $H(\mathbf{x})$:

$$\begin{aligned} H(\mathbf{x}) = &\sum_{t=0}^{T-1}\sum_{s\in\mathcal{S}_t} C_{\text{obs}}(s)x_{t,s} + \sum_{t=0}^{T-2}\sum_{s\in\mathcal{S}_t}\sum_{s'\in\mathcal{S}_{t+1}} C_{\text{mot}}(s,s')x_{t,s}x_{t+1,s'} \\ &+ \sum_{t=0}^{T-1}\lambda_{\text{eq}}\left(-\sum_{s\in\mathcal{S}_t} x_{t,s} + 2\sum_{\substack{s,s'\in\mathcal{S}_t\\ s<s'}} x_{t,s}x_{t,s'} + 1\right) \\ &+ \sum_{\text{invalid } (t,s,s')} \lambda_{\text{ineq}} x_{t,s}x_{t+1,s'} + \lambda_{\text{start}}(-x_{0,s_{\text{start}}} + 1) + \lambda_{\text{goal}}(-x_{T-1,s_{\text{goal}}} + 1). \end{aligned} \tag{7}$$

Step 5 — Q Matrix Entries and Offset. Collecting linear and quadratic coefficients gives the general closed-form expressions for the upper-triangular Q matrix and constant offset c:

$$Q_{(t,s),(t,s)} = C_{\text{obs}}(s) - \lambda_{\text{eq}} + \lambda_{\text{start}}\delta_{t,0}\delta_{s,s_{\text{start}}} + \lambda_{\text{goal}}\delta_{t,T-1}\delta_{s,s_{\text{goal}}}, \tag{8}$$

$$Q_{(t,s),(t+1,s')} = C_{\text{mot}}(s,s') + \lambda_{\text{ineq}} \cdot \mathbb{I}((s,s') \text{ is invalid}), \tag{9}$$

$$Q_{(t,s),(t,s')} = 2\lambda_{\text{eq}} \quad \text{for } s \neq s', \tag{10}$$

$$Q_{(t,s),(t',s')} = 0 \quad \text{otherwise}, \tag{11}$$

$$c = T\lambda_{\text{eq}} + \lambda_{\text{start}} + \lambda_{\text{goal}}. \tag{12}$$

The QUBO is then formulated as $\min_{\mathbf{x}\in\{0,1\}^n} \mathbf{x}^\top Q\mathbf{x} + c$.

Penalty Weight Justification. The penalty weights must dominate any potential reduction in the original objective function to enforce feasibility. Specifically, $\lambda_{\text{eq}} > \max_s C_{\text{obs}}(s) \cdot \max_t |\mathcal{S}_t|$ ensures that violating the one-state-per-step constraint costs more than the maximum possible decrease in obstacle costs. $\lambda_{\text{ineq}} > \max_{s,s'} C_{\text{mot}}(s,s')$ guarantees that the penalty for an invalid transition exceeds any motion cost savings. Finally, $\lambda_{\text{start}}, \lambda_{\text{goal}} > \max_s C_{\text{obs}}(s)$ ensures the start and goal states are strictly fixed.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^n$ recovers the optimal path because any feasible solution satisfies all constraints exactly, yielding zero penalty energy. Any infeasible solution incurs a penalty strictly greater than the maximum possible decrease in the original objective function (by the choice of λ parameters). Consequently, the global minimum of $H(\mathbf{x})$ cannot be infeasible; it must correspond to a feasible solution that minimizes the original cost function.

Illustrative Example. Consider a concrete instance with $T = 3$ time steps and $|\mathcal{S}_t| = 3$ states per step, labeled $\{0, 1, 2\}$. Variables are flattened to a single index $k = 3t + s$, yielding $n = 9$ variables $x_0, \ldots, x_8$. Assume uniform obstacle costs $C_{\text{obs}}(s) = c_o$ and uniform motion costs $C_{\text{mot}}(s, s') = c_m$. Suppose the only forbidden transition is $(s = 0, s' = 0)$ between $t = 0$ and $t = 1$, corresponding to variables x_0 and x_3. The start state is $s_{\text{start}} = 0$ (x_0) and the goal state is $s_{\text{goal}} = 2$ (x_8).

The concrete objective is:

$$\mathcal{O}(\mathbf{x}) = c_o \sum_{k=0}^{8} x_k + c_m(x_0x_3 + x_0x_4 + x_0x_5 + x_1x_3 + x_1x_4 + x_1x_5 + x_2x_4 + x_2x_5 + x_3x_6 + x_3x_7 + x_4x_6 + x_4x_{\cdots} \tag{13}$$

The concrete penalty terms expand to:

$$\mathcal{P}_{\text{eq}} = \lambda_{\text{eq}} \left(-\sum_{k=0}^{8} x_k + 2(x_0x_1 + x_0x_2 + x_1x_2 + x_3x_4 + x_3x_5 + x_4x_5 + x_6x_7 + x_6x_8 + x_7x_8) + 3 \right), \tag{14}$$

$$\mathcal{P}_{\text{ineq}} = \lambda_{\text{ineq}} x_0 x_3, \tag{15}$$

$$\mathcal{P}_{\text{start}} = \lambda_{\text{start}}(-x_0 + 1), \quad \mathcal{P}_{\text{goal}} = \lambda_{\text{goal}}(-x_8 + 1). \tag{16}$$

Combining these, the full Hamiltonian $H(\mathbf{x}) = \mathcal{O}(\mathbf{x}) + \mathcal{P}_{\text{eq}} + \mathcal{P}_{\text{ineq}} + \mathcal{P}_{\text{start}} + \mathcal{P}_{\text{goal}}$ collects coefficients for each monomial. The optimal feasible path selects states $s_0 = 0, s_1 = 1, s_2 = 2$, corresponding to the binary vector $\mathbf{x}^* = (1, 0, 0, 0, 1, 0, 0, 0, 1)$. Substituting $\mathbf{x}^*$ into $H(\mathbf{x})$ yields the minimum energy:

$$H(\mathbf{x}^*) = 3c_o + 2c_m. \tag{17}$$

This matches the sum of obstacle costs for three steps and motion costs for two transitions, confirming the formulation correctly encodes the instance.